

Designer Genes C - Designer Genes - Rickards Invitational Div. C SATELLITE - 11-01-2025
Student Answer Sheet (/rickards/ES/AnswerSheet?tid=000G7Z)

Instructions (shown before students start the test)

Welcome to RSOI 2026! Please remember that you are allotted:

1 8.5" x 11" physical double-sided cheatsheet

2 Class II calculators

Remember that event supervisors are busy and will most likely not be able to respond to questions. We apologize in advance for that

Good luck!

Written by: Eric Liu @ WMHS

Introduction (shown after students start the test)

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1. (3.00 pts)

Iron is a very important trace element in your body, notably used in the production of hemoglobin. However, it must be regulated closely to prevent conditions like hemochromatosis from occurring. Imagine in a system where intestinal cells have a genetic regulatory system that codes for ferritin and transferrin. Transferrin transports iron to cells in the body and ferritin sequesters iron and gets eventually lost in the intestine. Assume all mRNAs for these proteins have the iron regulatory sequence which is bound to by the iron response protein. This protein is deactivated by binding of iron. Use your discretion for regulatory mechanisms. Select all of the following that would make evolutionary sense. (or just select what is true)

(Mark **ALL** correct answers)

- ☐ A) The transferrin IRE is located on the 5' end of the transcript, allowing low iron to promote transcription.
- ☐ B) In an IRE knockout strain in high iron conditions, transferrin would be mostly produced.
- ☐ C) In an IRE knockout strain in low iron conditions, ferritin would be mostly produced.
- ☐ D) In an IRE knockout and a hyperactive IRP in low iron conditions, transferrin would be mostly produced.
- ☐ E) In a strain of mice with a hypomorph IRE, they would have a higher risk of hemochromatosis.

2. (3.00 pts) DNA replication and transcription in eukaryotes and prokaryotes have some major differences. Select all of the following that are true.

(Mark **ALL** correct answers)

- ☐ A) In prokaryotes, the primosome produces a primer made out of RNA and DNA, unlike the eukaryotic primosome which only makes a primer out of RNA.
- ☐ B) DNA ligase in prokaryotes is powered by ATP, while in eukaryotes and archaea, it is powered by NADPH.
- ☐ C) To remove Okazaki fragments, eukaryotes use DNA polymerase epsilon on the lagging strand to push off the primers and cut it with flap endonuclease.
- ☐ D) To remove Okazaki fragments, prokaryotes use DNA polymerase I to cut in the 3' to 5' direction and remove the Okazaki fragment, then replicating over the gap in DNA.
- ☐ E) RNA polymerase in prokaryotes and eukaryotes both use the beta subunits for catalytic activity, while the other subunits are for other functions.

3. (3.00 pts)

Nothing in biology makes sense except in the light of evolution. Everything (most stuff), from plasmalogenes in heart tissue to sustain it under heavy oxidative stress to the fine structure of a neuron has been shaped through evolution. Plasmids, which I hope you know by now, will also make sense in the light of evolution. Select all of the following that are true. (this question will make sense in the light of evolution)

(Mark **ALL** correct answers)

- ☐ A) F factors promote their own genes by containing genes that allow for their own replication and transmission into other bacteria, allowing their genes to be passed on through generations.
- ☐ B) Col plasmids kill all other bacteria around them including other col plasmid containing bacteria to make sure their host is able to proliferate and survive.
- ☐ C) Virulence plasmids turn their host pathogenic so they can outcompete and spread themselves.
- ☐ D) R plasmids provide antibiotic resistance to their bacterial host so they can become pathogenic and kill their host organism.
- ☐ E) Transposons are similar to plasmids in which they both may provide some benefit to their host, but as a side effect of making it so they are not selected against.

4. (3.00 pts)

At the oriC of a bacterial chromosome, there are three important sequences, being the 5 DnaA boxes, GATC methylation sites, and the AT rich region. Select all of the statements that are true.

(Mark **ALL** correct answers)

- ☐ A) The DnaA boxes need to bind to GTP bound DnaA proteins.
- ☐ B) The GATC methylation sites are required to be unmethylated for replication to occur to prevent excessive replication.
- ☐ C) In a mutant bacteria which has the AT rich region deviate from the consensus region, the time it takes for it to replicate decreases due to decreased attractions.
- ☐ D) Because S adenosyl methyltransferase has an electrophilic methyl group due to a positive sulfur, cytosine requires activation by DNA methyltransferase to act as a nucleophile.

- ☐ E) DnaA recruits a helicase to assist with unwinding the DNA strands.

5. (3.00 pts) The lactose operon is really funny. Zen Keng thinks he knows the lactose operon. Select all of the statements that are true.

(Mark **ALL** correct answers)

- ☐ A) When glucose metabolizing enzymes are shut down in a bacteria, the lactose operon will be active when exposed to lactose.
- ☐ B) If an adenyl cyclase inhibitor is added to a bacterial medium, the lactose operon will transcribe with low affinity.
- ☐ C) If a merozygote $lacI^{+/-}$ strain with $lacY^{-}$ and $lacI^{-}$ on the main chromosome had their genetic material combined with a strain with $lacP^{-}$, the lactose operon will be active.
- ☐ D) The affinity for the repressor to the operator site is higher in low glucose concentrations than high glucose concentrations.
- ☐ E) The repressor binds to all three operator sites, and removal of any will decrease the effect of inhibition.

6. (2.00 pts) Tautomerization is pretty bad. Select all of the following that are true.

(Mark **ALL** correct answers)

- ☐ A) Tautomerization involves the shifting of a double bond and a methyl group.
- ☐ B) One favorable tautomerization is a keto to enol group, forming a stable enol.
- ☐ C) Tautomeric base pairing rules are AG and TC.
- ☐ D) Tautomerization must happen during replication to cause mutations.

7. (1.00 pts) What is the structure of a zinc finger motif?

8. (1.00 pts) What part of a DNA helix does the zinc finger motif recognize?

9. (1.00 pts) What is the main role of leucine zippers in protein interactions?

10. (2.00 pts) How does cAMP cause change in gene regulation? Be specific about protein protein interactions and the normal state vs the activated state.

11. (2.00 pts) Histones are pretty cool. Select all of the following that are true.

(Mark **ALL** correct answers)

- ☐ A) H1.0 is a histone involved in histone compaction and gene repression.
- ☐ B) H2A.Z and H3 are the core histones found near transcriptional start sites.
- ☐ C) spH2B is the core histone in sperm cell DNA.
- ☐ D) macroH2A is found in X chromosomes.
- ☐ E) H3.3 is involved in gene repression.

12. (2.00 pts) Describe how CpG methylation decreases transcription.

13. (2.00 pts)

Oh no! You accidentally knock out the CAF1 protein in a mouse. Ignoring all the questions that raises, including how, why, and why isn't this mouse dead, explain how this is detrimental to the mouse for gene expression.

14. (5.00 pts) DNA pol III has a couple subunits. List the functions of the following subunits: alpha, epsilon, theta, beta, gamma.

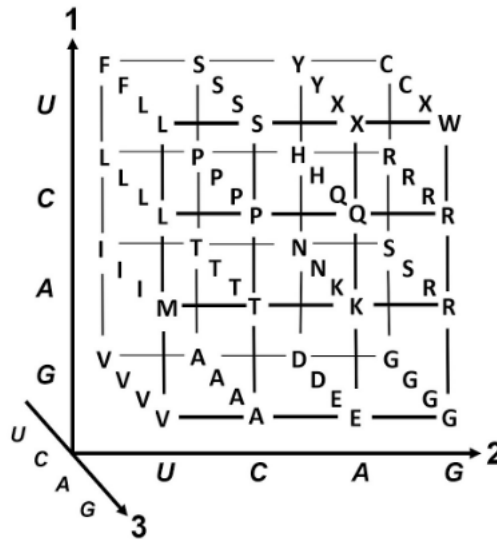
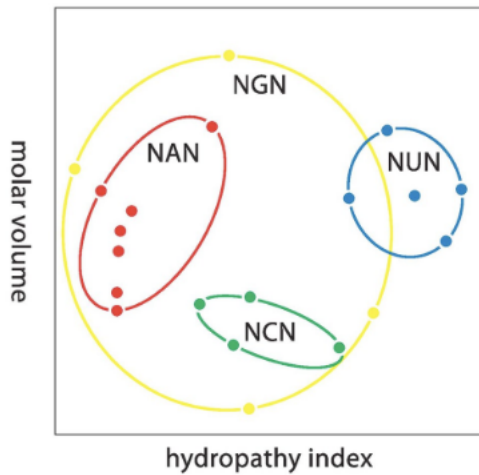
15. (1.00 pts)

You observe a mouse with increased DNA catenanes and you wonder why this is occurring. You find out that Zeng Ken knocked out a random gene in the mouse. What did this gene code?

16. (2.00 pts) What determines the origin of replication in animals (generally) and what determines the origin of replication in bacteria?

17. (3.00 pts) What does the origin replication complex do?

Codons are pretty cool. Note the following IUPAC ambiguity codes might be useful for this section. N = any base (A, U, G, or C), Y = a pyrimidine, U = a purine



18. (2.00 pts) Why does mutation at the third codon position often result in changes that do not affect protein structure?

19. (2.00 pts) If a mutation changes a codon from NUN to NCN, will the protein remain functional as normal? Assume that the change occurs at the 2nd position only.

20. (3.00 pts) Which codon position will cause the most “radical” changes when mutated? Justify your answer with data from the figures.

21. (5.00 pts)

Consider the following 2 amino acids: amino acid A is encoded by YUR/CUN and amino acid B is encoded by UCN/AGY. For each base position, determine the amino acid that is the most resistant to mutation (say equal they’re very similar/the same), and briefly justify your answer for each. Also, determine which amino acids A and B actually correspond to.

22. (3.00 pts) In terms of the graph with the hydropathy index, how might codons become biased in an organism that lives in very high temperatures? Explain.

23. (3.00 pts) Mutations at which codon positions can cause charge reversal of the amino acid encoded?

- ☐ A) 1st
- ☐ B) 2nd
- ☐ C) 3rd
- ☐ D) More than one of the above
- ☐ E) None of the above

24. (3.00 pts)

Assume that as a certain species is evolving, a single nucleotide base substitution in a codon occurs with equal probability in each codon. If the average molar volume difference between adjacent codons is 15, and across the entire cube it is 50, calculate the fraction of mutations that are conservative. A conservative mutation will be defined as having a molar volume change of ≤ 20 .

25. (3.00 pts) Using the same situation as before, how much does this code reduce mutation compared to one that is random?

26. (5.00 pts)

When crossing different strains of mice, you achieve the following table of phenotypes. There are 4 genes that control color. Based on the data, select all of the statements that are

true.

Cross	F1	F2
AaBbCCDDxAaBbCCDD	All blue	1B:1Y:2R
aaBBCCDDxaaBBccDd	All red	1B:1W:2R
aaBbccDDxaabbccDD	All blue	1B:1Y
aaBBccDDxaaBBccDd	All blue	1B:1W
aabbccddxaabbccdd	All white	All white

(Mark **ALL** correct answers)

- ☐ A) There is one synthetic pathway for the pigment.
- ☐ B) When all genes are functional, the only pigment produced is blue.
- ☐ C) Yellow is dominant to red.
- ☐ D) If you did a heterozygous cross, the chance of obtaining a yellow mouse is 15/64.
- ☐ E) The probability of obtaining a white mouse in a heterozygous cross is $\frac{1}{4}$

27. (5.00 pts)

Zen Keng is doing restriction digests on an unknown plasmid with restriction enzymes *ihateIV* and *microbemissionVII*. When run on the gel with a restriction enzyme with only one cutting site, Zen gets a band at 8.1kB. When run with *ihateIV* only, he gets two bands, one at 6.3kB and one at 0.9kB. When run with *microbemissionVII*, three bands appear at 5.6kB, 2.2kB, and 0.3kB. When cut with both, he gets a gel with one band at 3.3kB, one band at 2.2kB, one band at 0.9kB, one band at 0.3kB, and one band at 0.5kB. Select all the following statements that are true.

(Mark **ALL** correct answers)

- ☐ A) There are 2 *ihateIV* cut sites.
- ☐ B) There are 3 *microbemissionVII* cut sites.
- ☐ C) Only one of the *ihateIV* segments is cut by *microbemissionVII*
- ☐ D) These results are consistent with an error in the data.

28. (5.00 pts) Describe the plasmid by explaining the relative orientations of each cut site.

29. (3.00 pts)

A researcher investigates the chromosomal arrangement of four loci—A, B, C, and D—in *Neurospora crassa*, a model fungus. They perform the cross $a b c d \times A B C D$.

From 500 dissected asci, the loci A and B recombination data is collected. It is evident that 52% have only undergone one crossover. For the rest of the asci, $\frac{1}{4}$ of them have no parental combinations. From separate pairwise crosses, the following recombination frequencies are observed:

- A–C = 25%
- B–C = 15%

(Mark **ALL** correct answers)

- ☐ A) A and B are genetically linked.
- ☐ B) The recombination frequency between A and B is between 26% and 36%.
- ☐ C) The pairwise differences aren't consistent with the ordering A–C–B due to slight discrepancies in recombination frequencies.
- ☐ D) The meiotic divisions undergone by the spores of *N. crassa* cannot be inferred by their spatial positions relative to other spores, as *N. crassa* asci are arranged circularly.
- ☐ E) Consider locus A and locus D. A 2-strand double crossover tends to produce an ascus with no PD, a 3-strand double crossover tends to produce an ascus with 4 PD, and a 4-strand double crossover tends to produce an ascus with all PD.

30. (3.00 pts) Which of the following are characteristics of the centromeres on chromosomes?

- ☐ A) Bacteria contains many modified histones on their centromeres of chromosomes
- ☐ B) Histones are acetylated with lysine at centromeres.
- ☐ C) Centromeres are associated with coding DNA.
- ☐ D) H3 Histones are replaced by CENP-A at centromeres.
- ☐ E) Centromeres are destroyed via nuclease activity to trigger the apoptotic cascade.

31. (3.00 pts) What are the 4 properties of genetic material that allow it to serve its function?

32. (3.00 pts) What type of bond is formed between nucleotide bases and its sugar called?

33. (3.00 pts) Describe the structure of B and Z DNA and list the conditions where Z DNA formation would be favorable.

34. (1.00 pts) What are the pairing rules for triplex bonding (AT and GC)?

35. (1.00 pts) Describe the general mechanisms by which topoisomerases relax supercoiling.

36. (2.00 pts) Describe the difference between topoisomerase IA/IB and IIA in terms of function and mechanism of action.

37. (1.00 pts) What is the difference between point and regional centromeres?

38. (2.00 pts) What is the difference between the solenoid and zigzag packing model for histones?

39. (2.00 pts)

How do RNA primers get removed in eukaryotes and how is this different from the RNA primer removal in prokaryotes? What if RNA primers are very long in eukaryotes?

40. (2.00 pts) What does MCM helicase do?

41. (2.00 pts) What happens during polymerase switch and would you expect this to happen on the leading or the lagging strand?

42. (2.00 pts) Describe the functions of each subunit in prokaryotic RNA polymerase.

43. (2.00 pts) What RNA polymerase subunit recognizes promoter sequences and what motif allows it to?

44. (3.00 pts) Describe the difference between rho dependent termination and rho independent termination and the different DNA sequences that they rely on.

45. (2.00 pts) What is the difference between the three main eukaryotic polymerases in terms of what they transcribe, and which is the most active?

46. (2.00 pts) What is the order of transcription factor binding to the promoter and what is the function of each one?

47. (2.00 pts) How does the mediator protein affect RNA polymerase, and how does this differ between enhancers and silences?

48. (2.00 pts) Define processing of RNA and give two examples of it.

49. (3.00 pts) Define polyA tailing of RNA, how it happens in eukaryotes, and when it occurs in bacteria.

50. (3.00 pts) Describe the mechanisms of action of group I and II introns and describe the role of maturases in this type of splicing.

51. (3.00 pts) How many subunits are in the spliceosome? What is the function of each one, and which ones do the catalytic processing that splices the RNA?

52. (3.00 pts) Describe the process of capping of RNA. Give ALL valid examples of this happening.

53. (1.00 pts) What is the start codon and what amino acid can it also code for?

54. (1.00 pts) What two codons code for selenocysteine and pyrrolysine respectively? How do ribosomes know when to insert selenocysteine?

55. (2.00 pts)

Cade Cunningham is investigating the cis regulation of a gene, which would inhibit it. He creates a merozygote, where the plasmid has the cis regulatory element and the genome has the knockout version. He finds out that the gene is not expressed. Is this consistent with the type of regulation and why?

56. (3.00 pts)

Cade Cunningham then decides to synthesize the protein product of this gene with cell free protein synthesis. Name 3 materials other than ribosomes, mRNA, and excess tRNA (with amino acids attached) that are needed for this system to function correctly.

57. (3.00 pts) His mRNA is 963 nucleotides long. How much GTP is needed to translate this protein fully from start to finish?

58. (1.00 pts) Give an example of an organic chemical used to extract DNA from samples.

59. (2.00 pts) Provide two advantages and two disadvantages of using organic chemicals in extraction of DNA from samples compared to other extraction methods.

60. (3.00 pts) What is the protocol for northern blotting? Give general steps, answers CAN vary.

61. (2.00 pts) Why is northern blotting no longer used?

62. (2.00 pts) How does RNA degradation differ from DNA degradation and what functional group explains this phenomenon?

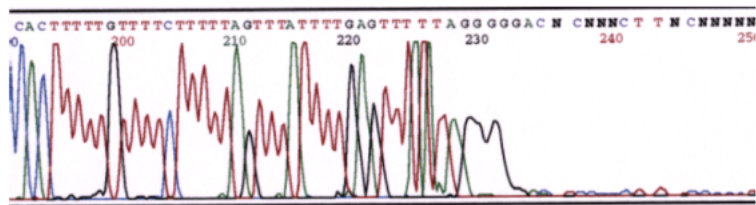
63. (2.00 pts) Select ALL the following that are true regarding DNA sequencing technology

(Mark ALL correct answers)

- ☐ A) Sanger sequencing produces longer reads than Illumina
- ☐ B) Capillary electrophoresis in Sanger sequencing is unable to distinguish between DNA molecules at the single-nucleotide resolution
- ☐ C) Sanger sequencing uses Taq Polymerase
- ☐ D) Sanger sequencing is more suitable for smaller DNA fragments than Illumina

64. (2.00 pts) Explain how the structure of ddNTPs allow them to function as “chain-terminating nucleotides” in Sanger sequencing.

65. (3.00 pts) The following chromatogram is obtained from a failed Sanger sequencing run. Based on the image, what is the cause of this failure? What tells you this?



66. (3.00 pts)

Yo croski, we fam in the 6ix, but answer my mans biotechnology question, bruv, or you a gerbert, my word. Which one of the following experimental questions can PCR or a PCR-utilizing technique NOT answer:

(Mark **ALL** correct answers)

- ☐ A) What is the change in expression of the Crodie gene after exposure to ethanol?
- ☐ B) What miRNAs regulate expression of the Crodie gene?
- ☐ C) What miRNAs are present in cancer cells after Crodie knockout?
- ☐ D) What is the methylation level of the Crodie gene in hepatocytes?

67. (3.00 pts) Order the following steps of illumina sequencing in order using numbers 1-5:

DNA denaturation
Laser recording and emission detection
Bridge amplification
Addition of dNTPs and fluorophore
Adaptor attachment

68. (3.00 pts) Explain the technique of tagmentation to generation DNA libraries for Next-Generation sequencing. What advantage does it have over traditional sonication?**69. (3.00 pts)**

John wants to know the change in expression of the gene SSBSG in a line of HeLa cells after treatment with the drug Employmentone. Therefore, he decides to run a RT-qPCR. When he looks at his results graph on his samples, he sees that his Ct value is above 30. Select all the following that could NOT have caused this result.

(Mark **ALL** correct answers)

- ☐ A) John used SYBR Green in the probe, which can bind to primer-dimers
- ☐ B) John forgot to add DMSO

- ☐ C) John forgot to add DNase to his mixture
- ☐ D) Too high dNTPs levels

70. (2.00 pts)

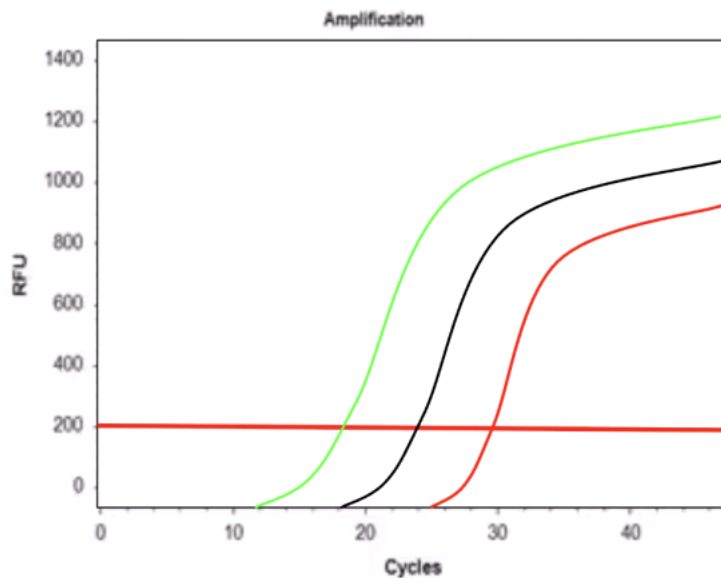
John reads a little more about PCR and RT-qPCR in order to troubleshoot. Which of the following is (are) correct statements regarding RT-qPCR and PCR in general.

(Mark **ALL** correct answers)

- ☐ A) PCR utilizes single-stranded DNA primers complementary to the 5' ends
- ☐ B) The RT in RT-qPCR stands for reverse-transcriptase.
- ☐ C) Pfu DNA polymerase is advantageous because it has 3'→5' exonuclease activity
- ☐ D) Taqman probe quencher activity increases as more DNA synthesis happens

71. (2.00 pts)

John reruns his samples using RT-qPCR and gets the following graph. The green represents the relative fluorescence units (RFU) from HeLa cells without treatment of Employmentone. The black represents the RFU of a housekeeping gene. The red represents the RFU from HeLa cells with treatment of Employmentone.



What is the purpose of the housekeeping gene in this study?

72. (2.00 pts) Using the $\Delta\Delta C_t$ method, find the change in gene expression in RFU. Show all work for full credit.

73. (2.00 pts) Conclude whether Employmentone increases or decreases expression of SSBSG and justify using your answer to the previous question.

Congratulations for finishing! Good luck with your other events!